

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1198

Sunnyslope 100 kV
January 28, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1198. The Interconnection Customer proposes a 188.5 MW solar photovoltaic facility interconnecting at the Sunnyslope 100 kV substation in the Ohio area. The request was received on May 01, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 347 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,489,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Drouen to Alps 100.0 kV
Contingency: Barrett to Norwich 500.0 kV
Loading: 185.1% (Rating: 127 MVA)
DFAX: 5.8%
Pre-project loading: 176.5%
2. Facility: Duranzo to Derouen 100.0 kV
Contingency: Barrett to Norwich 500.0 kV L
Loading: 182.9% (Rating: 114 MVA)
DFAX: 5.8%
Pre-project loading: 173.2%
3. Facility: Carencro to Sunnyslope 100.0 kV
Contingency: Wolfe Substation to Brrett 500.0 kV line
Loading: 169.1% (Rating: 124 MVA)
DFAX: 58.2%
Pre-project loading: 80.3%
4. Facility: Carencro to Devon 100.0 kV L
Contingency: Wolfe to Barrett 500.0 kV
Loading: 168.6% (Rating: 113 MVA)
DFAX: 58.2%
Pre-project loading: 71.1%
5. Facility: Star City to Bellvue 100.0 kV Cir. 2
Contingency: Star City to Blvue 100.0 kV Ckt 1
Loading: 130.1% (Rating: 160 MVA)
DFAX: 5.9%
Pre-project loading: 123.2%

6. Facility: Devon to Delta York Ec Substation 100.0 kV
Contingency: Wolfe to Barrett 500.0 kV
Loading: 129.1% (Rating: 122 MVA)
DFAX: 58.2%
Pre-project loading: 39.1%
7. Facility: Poncatoula Substation to Sunnyslope Substation 100.0 kV
Contingency: Sunnyslope to Carencro 100.0 kV
Loading: 128.0% (Rating: 132 MVA)
DFAX: 100.0%
Pre-project loading: 14.6%
8. Facility: Star City to Bellvue 100.0 kV Cir. 1
Contingency: Star City to Bellvue 100.0 kV
Loading: 125.7% (Rating: 166 MVA)
DFAX: 5.9%
Pre-project loading: 119.0%
9. Facility: Edgewater Substation to Craigmont 100.0 kV line
Contingency: Barrett to Norwich 500.0 kV
Loading: 108.2% (Rating: 128 MVA)
DFAX: 8.1%
Pre-project loading: 96.2%
10. Facility: Federal Hts to Ltg Pleasant Hill 100.0 kV Line
Contingency: Barrett to Norwich 500.0 kV
Loading: 104.8% (Rating: 335 MVA)
DFAX: 6.8%
Pre-project loading: 100.9%
11. Facility: Craigmont Substation to Ltg Pleasant Hill 100.0 kV line
Contingency: Norwich to Barrett 500.0 kV
Loading: 104.5% (Rating: 351 MVA)
DFAX: 6.8%
Pre-project loading: 100.9%
12. Facility: Windmill to Federal Heights 100.0 kV
Contingency: Barrett to Norwich 500.0 kV Line
Loading: 100.5% (Rating: 135 MVA)
DFAX: 5.0%
Pre-project loading: 93.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alps to Derouen 100.0 kV	Rebuild line with higher-capacity conductor	\$1,547,000	100% (\$1,547,000)
Duranzo to Derouen 100.0 kV	Rebuild line with higher-capacity conductor	\$3,130,000	100% (\$3,130,000)
Sunnyslope to Carencro 100.0 kV	Rebuild line with higher-capacity conductor	\$1,710,000	100% (\$1,710,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Devon to Carencro 100.0 kV	Rebuild line with higher-capacity conductor	\$1,390,000	100% (\$1,390,000)
Star City to Bellvue 100.0 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,320,000	100% (\$1,320,000)
Devon to Delta York Ec Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$1,966,000	100% (\$1,966,000)
Poncatoula Substation to Sunnyslope Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$2,542,000	100% (\$2,542,000)
Bellvue to Star City 100.0 kV	Rebuild line with higher-capacity conductor	\$1,994,000	100% (\$1,994,000)
Craigmont to Edgewater 100.0 kV Line	Reconductor line	\$510,000	100% (\$510,000)
Federal Heights to Ltg Pleasant Hill 100.0 kV	Reconductor line	\$589,000	36% (\$211,000)
Craigmont Substation to Ltg Pleasant Hill 100.0 kV line	Reconductor line	\$631,000	36% (\$226,000)
Windmill to Federal Heihts 100.0 kV	Reconductor line	\$943,000	100% (\$943,000)
	Total allocated to project		\$17,489,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.